DECIMALS

Grade 4

COMPARING DECIMALS IN REAL-LIFE

When do comparing decimals come into play in your daily adventures?

OUR LEARNING TARGET

I can understand, compare and work with decimals in real - life situations.

Decimal Place Value Chart

Thousands	Hundreds	Tens	Ones	tenths	hundredths	thousandths
Th	H	T	O	t	h	th

SYMBOLS OF THE DAY

Before we dive in, let's recall what these symbols mean.

GREATER THAN

Used when the first number has a larger value than the second number.

$5 > 3$

LESS THAN

Used when the first number has a smaller value than the second number.

$3 < 5$

EQUAL

Used when the first number has the same value as the second number.

$5 = 5$

COMPARING USING PLACE VALUE CHARTS

STEP 1 Place the digits in their corresponding place value.

Another term to describe the ones place is units.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 2 Compare the digits from the largest place value.

A. If the digits are the same, move to the next place value to the right.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 2 Compare the digits from the largest place value.

B. If the digits are different, determine which digit is greater.

units	.	tenths	hundredths
5	.	8	7

units	.	tenths	hundredths
5	.	3	9

COMPARING USING PLACE VALUE CHARTS

STEP 3 Use the symbols, >, <, or = to describe the relationship of two numbers.

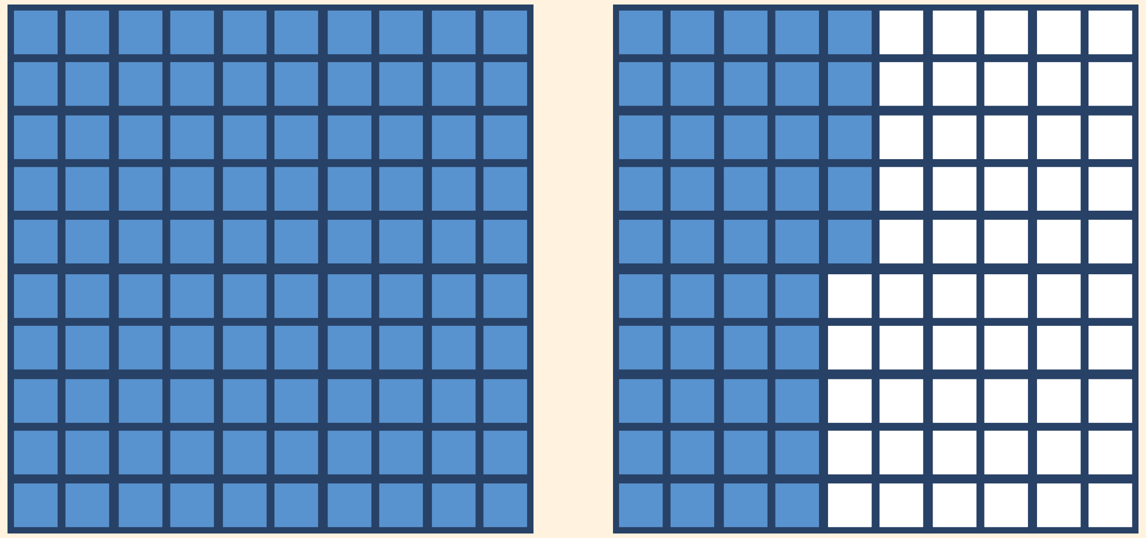
units	.	tenths	hundredths
5	.	8	7
5	.	3	9

In the tenths place, 8 is greater than 3. This means that 5.87 is greater than 5.39. In symbols, 5.87 > 5.39.

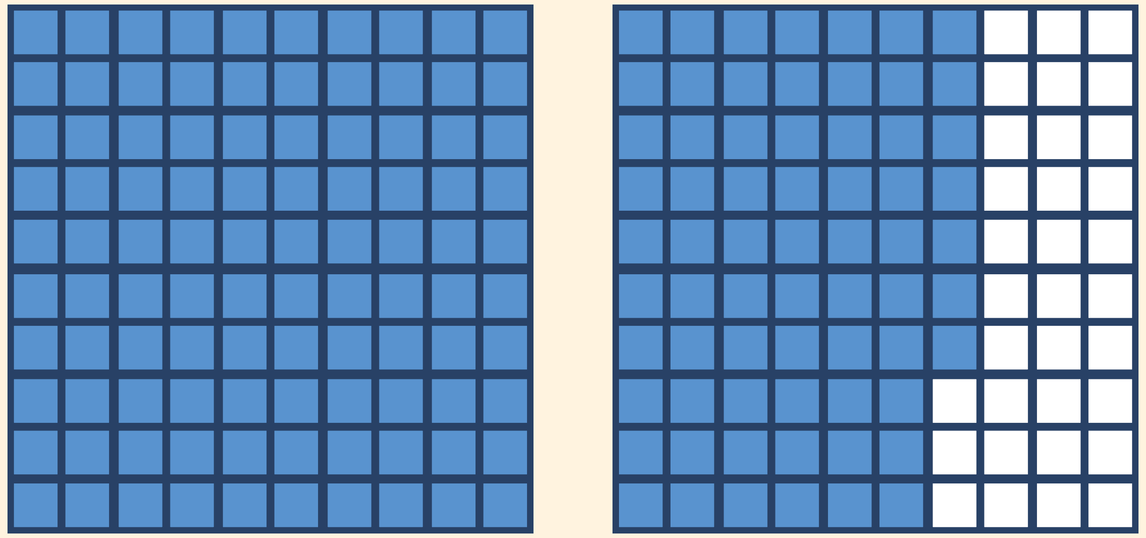
COMPARING USING AREA MODELS

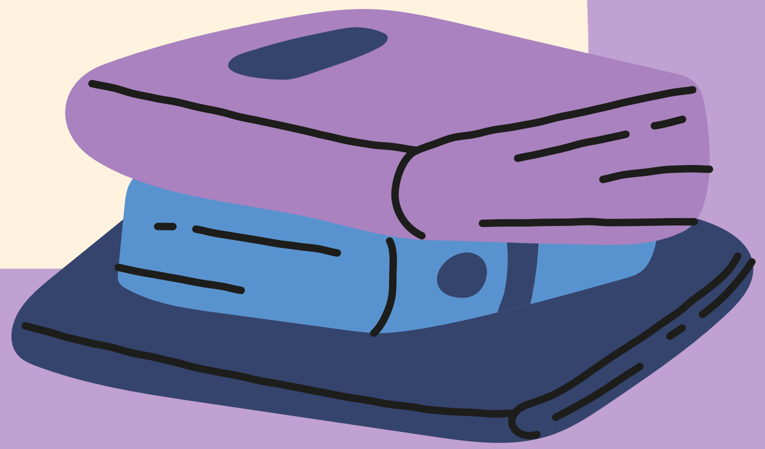
STEP 1 Draw a model for each decimal. Remember that 0.01 is 1 out of 100 parts and 1 is equivalent to 100 parts.

1.45



1.67

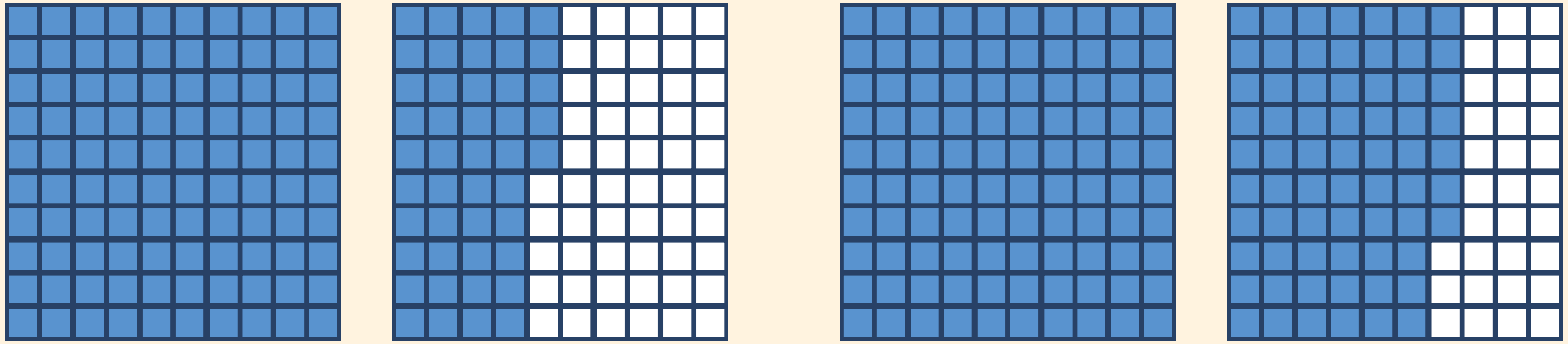




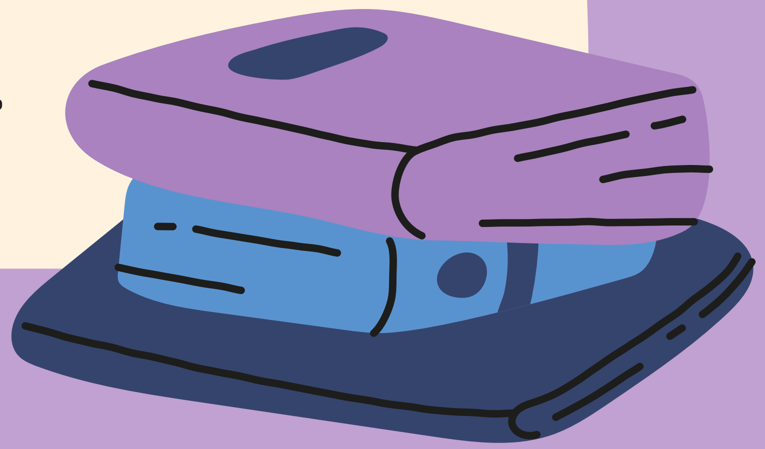
COMPARING USING AREA MODELS

STEP 2 Compare the shaded regions of each model. The number with more shaded regions has greater value.

1.45 1.67

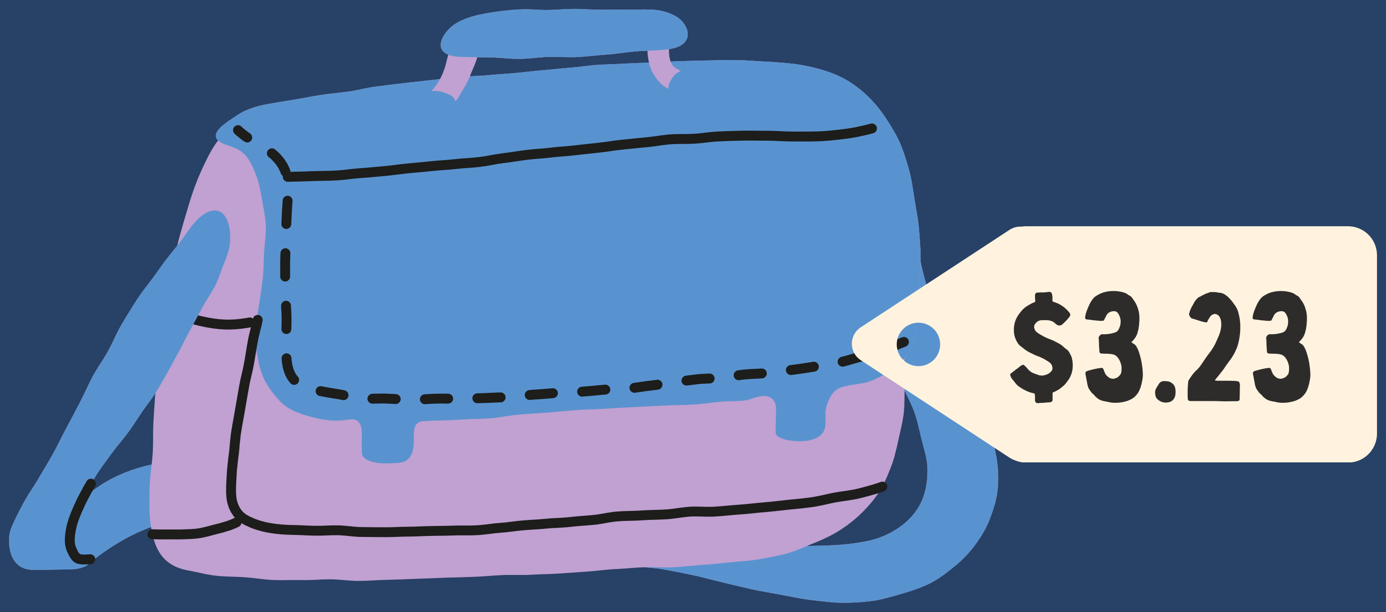


The number with more shaded regions has greater value.



WHICH BAG COMES WITH A LOWER PRICE TAG?

Place <, >, = on the blank to make the statement true.



ANSWER KEY

WHICH BAG COMES WITH A LOWER PRICE TAG?

Place <, >, = on the blank to make the statement true.

units	.	tenths	hundredths
3	.	2	6
3	.	2	3

$>$

The satchel comes with a lower price tag.

Adding and Subtracting Decimals to Hundredths

Learn how to perform operations on decimals

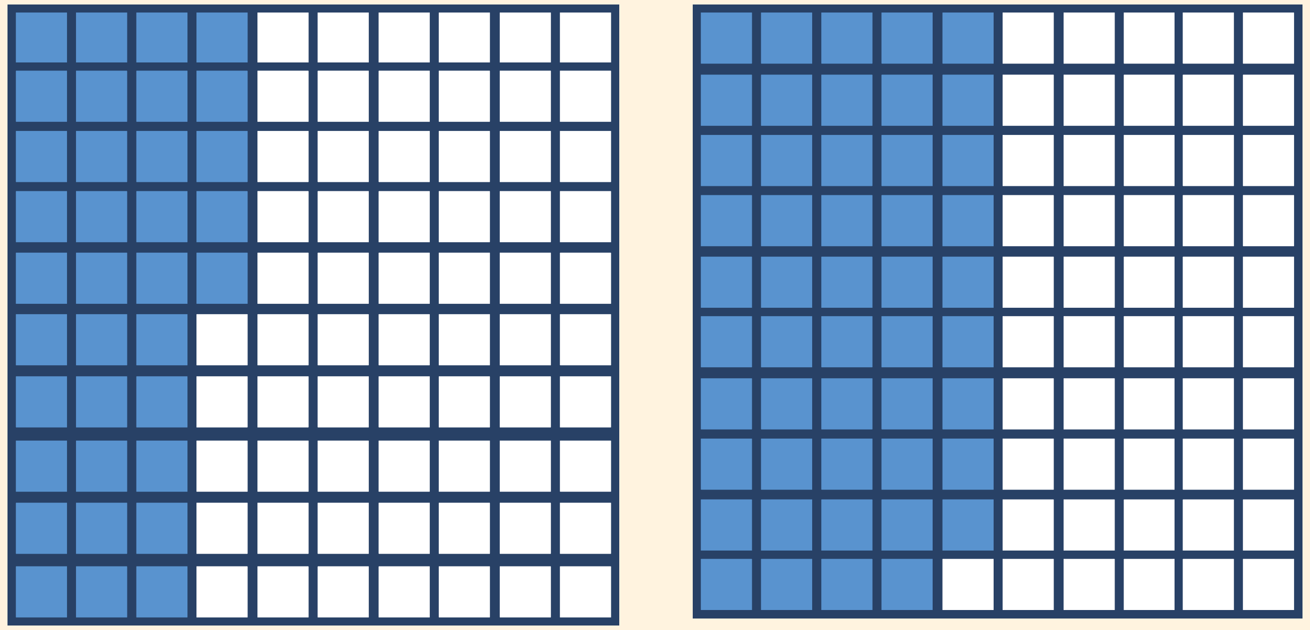
RECAP

We can compare two decimal numbers using the following methods.

PLACE VALUE CHART

units	.	tenths	hundredths
3	.	2	6
3	.	2	3

AREA MODELS



Adding and Subtracting Decimals to Hundredths

Learn how to perform operations on decimals

INSPIRED BY mathspace

Add decimals to hundredths

Just like adding whole numbers, we can add decimals by lining up the digits based on their place value.

Let's add 2.18 and 1.31.

	Units	.	Tenths	Hundredths
	2	.	1	8
+	1	.	3	1

TIP: When trying to line up the values, you can line up the decimal symbol first.

	Units	.	Tenths	Hundredths
	2	.	1	8
+	1	.	3	1
	3	.	4	9

The sum of 2.18 and 1.31 is **3.49.**

After lining up the digits of 2.18 and 1.31 based on place value, we can now add from right to left.

Start by adding the values on the Hundredths place, then the Tenths place, and finally the Units place.

Remember to write the decimal symbol in the sum!

Let us try another example!

Add 3.46 and 2.72.

	Units	.	Tenths	Hundredths
	3	.	4	6
+	2	.	7	2
			11	8

But wait! The sum of 4-tenths and 7-tenths is 11-tenths. When we get a two-digit sum, we need to perform **regrouping.**

We will regroup 10-tenths as 1 whole number or 1 unit.

	Units	.	Tenths	Hundredths
	1			
	3	.	4	6
+	2	.	7	2
	6	.	1	8

The sum of 3.46 and 2.72 is **6.18.**

Subtract decimals to hundredths

Just like in adding decimals, we follow the same method in subtraction.

Let us do 8.74 - 2.5.

	Units	.	Tenths	Hundredths
	8	.	7	4
-	2	.	5	0
	6	.	2	4

8.74 - 2.5 = **6.24**

TIP: After lining up the digits in their place values, any empty space can be filled in with zeroes.

2.5 is exactly the same as 2.50.

Let us try 23.15 - 10.42.

Tens	Units	.	Tenths	Hundredths
2	3	.	1	5
- 1	0	.	4	2
				3

But wait! How do we subtract 4-tenths from 1-tenths if it is bigger? Since we cannot subtract directly in the tenths column, we'll do **regrouping** of 1 from the units column.

Let us try 23.15 - 10.42.

	Tens	Units	.	Tenths	Hundredths
	2	~~3~~ 2	.	~~1~~ 11	5
-	1	0	.	4	2
					3

Take away 1 from the units place, and it will be added to the tenths place as 10-tenths.

	Tens	Units	.	Tenths	Hundredths
	2	2	.	11	5
-	1	0	.	4	2
	1	2	.	7	3

23.15 - 10.42 = **12.73**

Summary

We can add or subtract decimals to hundredths place with or without regrouping.

Keep in mind the following:

- Always line up the digits of the decimals based on their place value.
- Always add or subtract from right to left, starting with the last and smallest place value.
- Remember to correctly include the decimal symbol in writing your final answer.

Add decimals to hundredths

Just like adding whole numbers, we can add decimals by lining up the digits based on their place value.

Let's add 2.18 and 1.31.

	Units	.	Tenths	Hundredths
	2	.	1	8
+	1	.	3	1

TIP: When trying to line up the values, you can line up the decimal symbol first.

Multiplying and Dividing Decimals

0.8×10

Answer: 8

$$0.8 \div 10$$

Answer: 0.08


Thank you
for
listening!